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%SPECTRAL DECOMPOSITION AND ANALYSIS OF COMPLEX SIGNALS
clc;
close all;
clear;

%construction
% Parameters
F1= 10;
F2=20;
F3=30;
A1= 10;
A2=5;
A3=20;
fs = 1000;

Ts = 1/fs;
t = -1:Ts:1;

% Signals generation
X1=A1*exp(1i*2*pi*F1*t);
X2=A2*exp(1i*2*pi*F2*t);
X3=A3*exp(1i*2*pi*F3*t);

x=X1 + X2 +X3;

% Signal components
x_real = real(x);
x_imaginary = imag(x);
x_mag = abs(x);
X_phase = angle(x);

% Plotting
figure;

subplot(3,3,1)
plot(t,x_real);
xlabel('Time (s)');
ylabel('Amplitude');
title('Real Part');
grid on;

subplot(3,3,2)
plot(t,x_imaginary);
xlabel('Time (s)');
ylabel('Amplitude');
title('Imaginary Part');
grid on;

subplot(3,3,3)
plot(t,x_mag);
xlabel('Time (s)');
ylabel('Magnitude');
```

```
title('Magnitude');
grid on;

subplot(3,3,4)
plot(t,X_phase);
xlabel('Time (s)');
ylabel('Phase (rad)');
title('Phase');
grid on;

%applying fft

n = length(x);

df = fs/n;
f = (0:n-1)*df;

y = fft(x);
y_mag=abs(y)/n;

subplot(3,3,5)
stem(f,y_mag);
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('FFT Magnitude Spectrum');
grid on;
xlim([0 35]);

% shifting to -ve

y_shift=fftshift(y);
y_shift_mag=abs(y_shift)/n;
f_shift = (-floor(n/2):ceil(n/2)-1)*df;

subplot(3,3,6)
stem(f_shift,y_shift_mag);
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('FFT Magnitude Spectrum');
grid on;
xlim([-35 35]);

%to find phase of shifted fft

y_shift_phase=angle(y_shift);

subplot(3,3,7)
stem(f_shift, y_shift_phase);
xlabel('Frequency (Hz)');
ylabel('Phase (rad)');
title('FFT Phase Spectrum');
grid on;
xlim([-35 35]);
```

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% Calculate PSD using Welch method

[pxx, f_psd] = pwelch(x, [], [], [], fs, 'centered');

subplot(3,3,8)
plot(f_psd, pxx);
grid on;

xlabel('Frequency (Hz)');
ylabel('Power/Frequency (w/Hz)');
title('Power Spectral Density of Signal');
xlim([-35 35]);

% Find dominant frequency components

[peaks, locations] = findpeaks(y_mag, f, ...
    'MinPeakHeight', 0.5, ...
    'MinPeakDistance', 5);

% Display dominant frequencies

disp('Dominant Frequencies (Hz):');
disp(locations);

% Display magnitude of dominant frequencies

disp('Magnitude of Dominant Frequencies:');
disp(peaks);

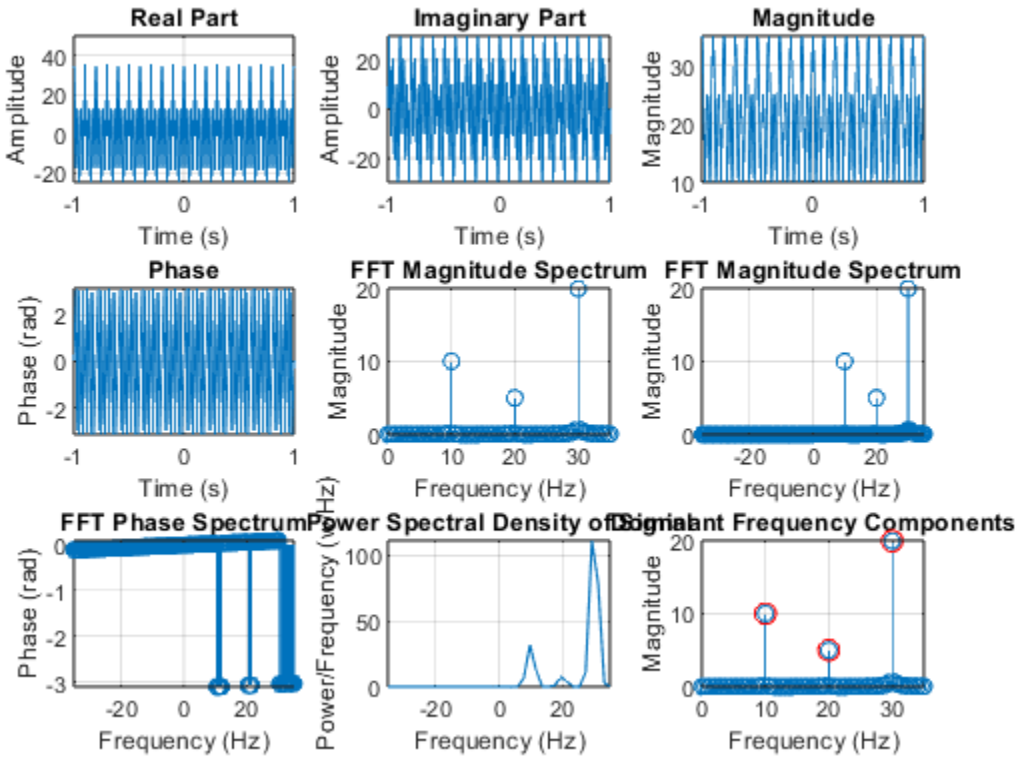
% Plot dominant frequencies

subplot(3,3,9)
stem(f, y_mag);
hold on;
plot(locations, peaks, 'ro', 'MarkerSize', 8);
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('Dominant Frequency Components');
grid on;
xlim([0 35]);
hold off;

% thanks
```

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Dominant Frequencies (Hz):
    9.9950   19.9900   29.9850

Magnitude of Dominant Frequencies:
   10.0183    5.0216   19.9629
```



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